

NI TB Genomic Surveillance

Outbreak Investigation Report

Generated: 2026-05-08 14:45:46 UTC

Case Summary

Total Cases	400
Clustered Cases	271
Unclustered Cases	129
Open Clusters	0

Automated Interpretation Flags

- Sequencing coverage is below target: 0.0% (target $\geq 80\%$).
- High-confidence transmission links present; prioritize epidemiology follow-up.

Analysis Summary

Posterior Samples	2000
MCMC Iterations	2500
Burn-in	500
Mean Log-Likelihood	-5250.50
Log-Likelihood SD	145.30
Transmission Probability	0.085
Generation Time Mean (days)	12.50
Generation Time SD (days)	3.200
Sampling Probability	0.650
Convergence Diagnostic	Rhat < 1.1 (good convergence)

Programme Surveillance KPIs (Last 12 Weeks)

Eligible Cases	53
Sequenced Cases	0
Sequencing Coverage (%)	0.0
QC Reported Cases	0
QC Pass Cases	0
QC Fail Cases	0
QC Pass Rate (%)	None
Contamination Flags	0
Median Days Specimen to QC	None

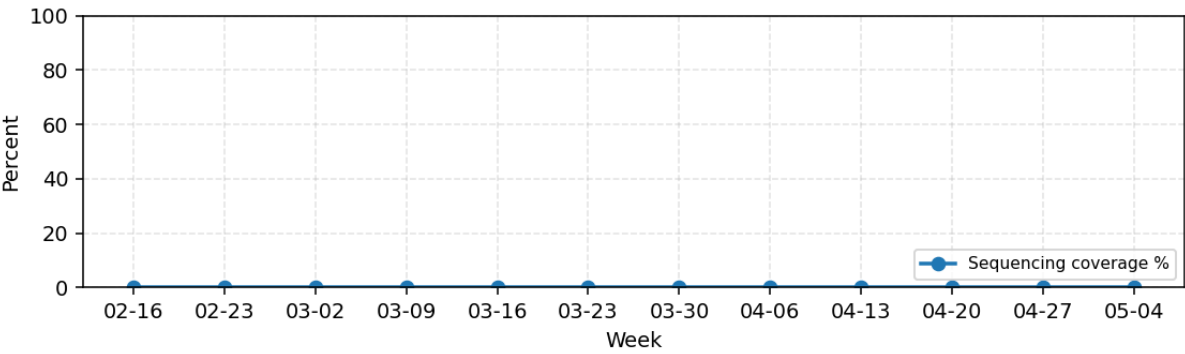
KPI Warning: sample_qc_metrics table not found; apply updated db/schema.sql to enable QC KPIs.

Regional Sequencing Representativeness

Region	Eligible	Sequenced	Coverage %
United Kingdom	53	0	0.0

Weekly Surveillance Trends (12 Weeks)

Week	Eligible	Sequenced	Coverage %	QC Pass %
2026-02-16	6	0	0.00	None
2026-02-23	5	0	0.00	None
2026-03-02	5	0	0.00	None
2026-03-09	5	0	0.00	None
2026-03-16	4	0	0.00	None
2026-03-23	3	0	0.00	None
2026-03-30	3	0	0.00	None
2026-04-06	7	0	0.00	None
2026-04-13	4	0	0.00	None
2026-04-20	4	0	0.00	None
2026-04-27	5	0	0.00	None
2026-05-04	2	0	0.00	None



Cluster Action Prioritization

Cluster	Cases	Regions	Most Recent	Status	Priority
e8289609	37	1	2026-04-28	monitoring	80
dea0bce5	32	1	2026-04-29	closed	70
5488a7ca	30	1	2026-05-02	monitoring	66
c86d1a07	29	1	2026-05-06	monitoring	64
07a3bd0a	27	1	2026-04-09	monitoring	59
e9837773	26	1	2026-04-26	monitoring	58
9a7b68f7	25	1	2026-04-23	monitoring	55
be74f343	24	1	2026-03-21	closed	52
0ce5ba2a	24	1	2026-03-14	monitoring	52
0ea960a7	17	1	2026-01-27	closed	37

Priority score combines cluster size, cross-region spread, specimen recency, and open investigation status.

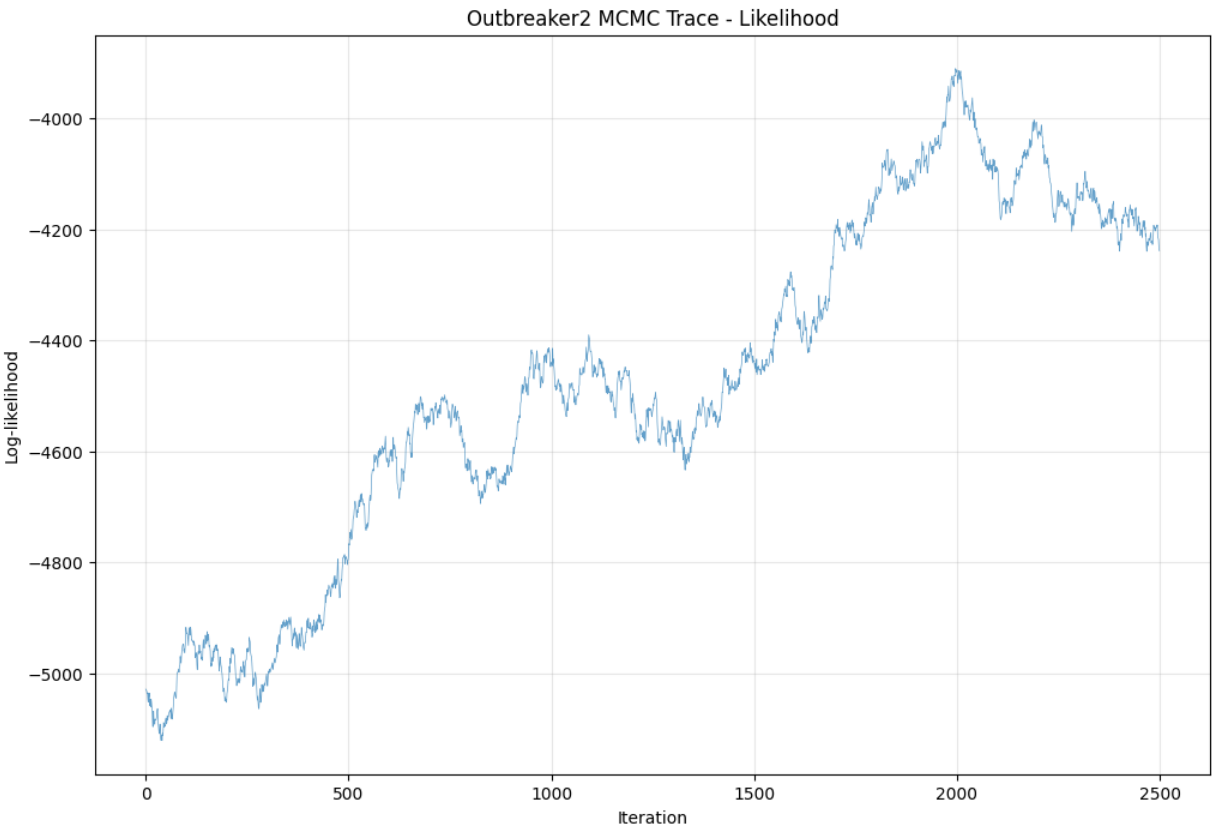
Transmission Priority Signals

Case	Region	Risk	Out	In
e2b2f0f6	United Kingdom	1.643	2	2
80a225c1	United Kingdom	1.622	2	2
786a97aa	United Kingdom	1.622	2	2
111b4914	United Kingdom	1.622	2	2
b759cfb0	United Kingdom	1.622	2	2
07ad8818	United Kingdom	1.621	2	2
4356cc32	United Kingdom	1.621	2	2
c7654356	United Kingdom	1.62	2	2
2a4d1b7e	United Kingdom	1.611	2	2
8be14a47	United Kingdom	1.608	2	2

Network snapshot: nodes=271, edges=512, high-confidence links=193

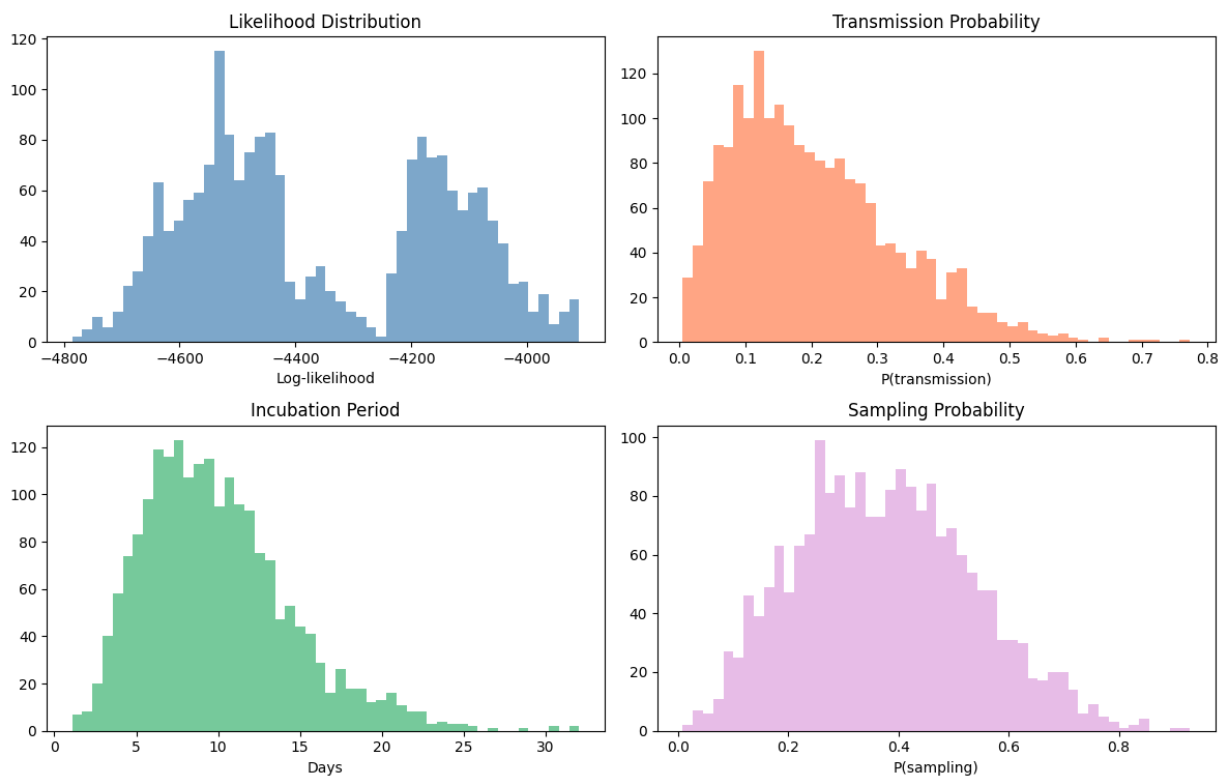
Diagnostic Graphics

Trace



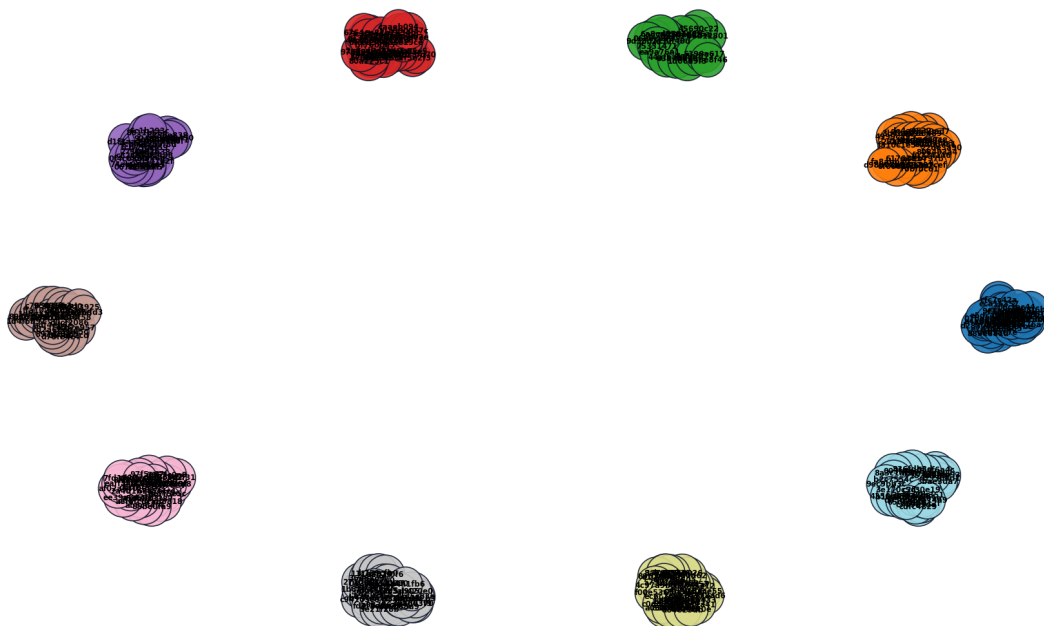
Hist

Outbreaker2 Posterior Distributions

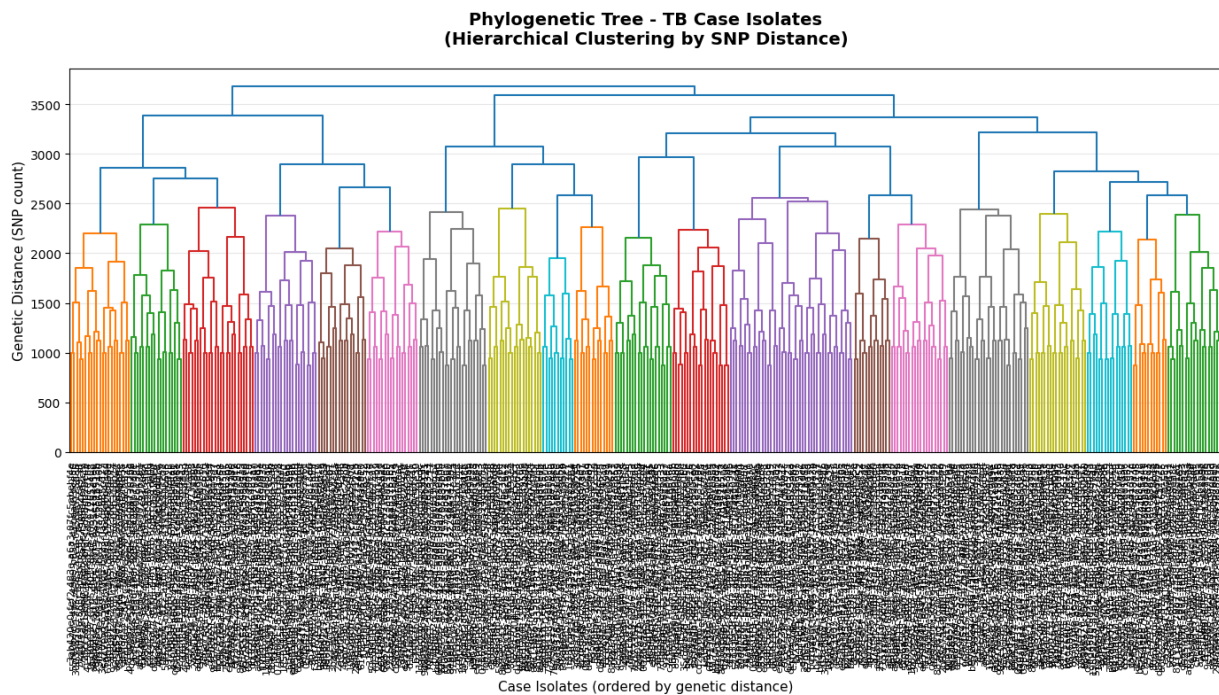


Tree

Enhanced Transmission Network
Node size = inferred spread risk | Edge label = transmission probability | Color = cluster



Phylo



Resistance



Data Provenance

This report combines case tables, cluster assignments, outbreaker summary metrics, surveillance KPIs, and transmission network outputs available at generation time.

Outbreaker summary timestamp: 2026-05-08T15:43:30.788998
Transmission network timestamp: 2026-05-08T14:43:35